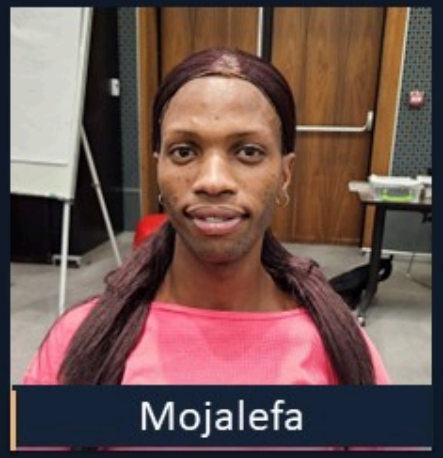
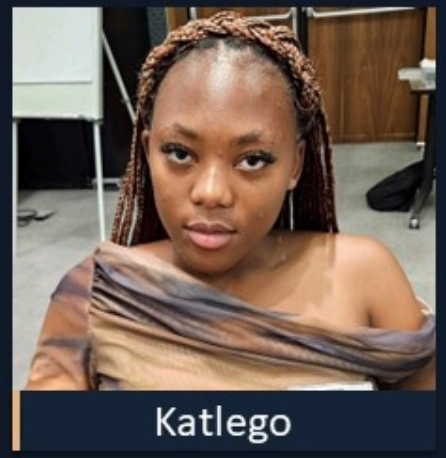
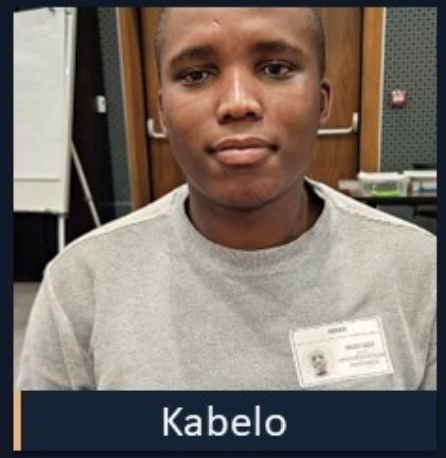
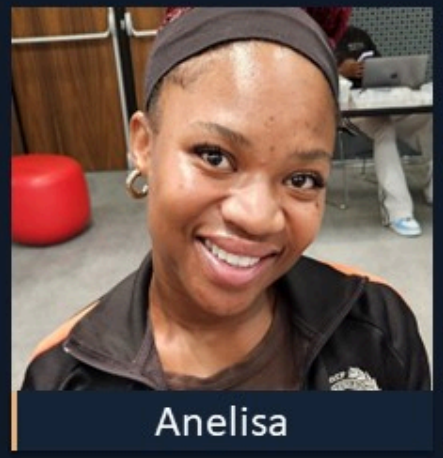
WESTRAND MASTERS

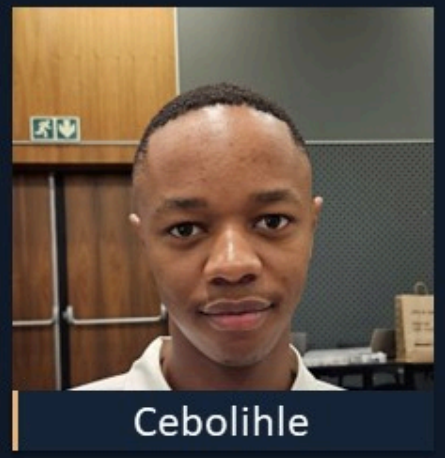
Pioneers of SA Educate

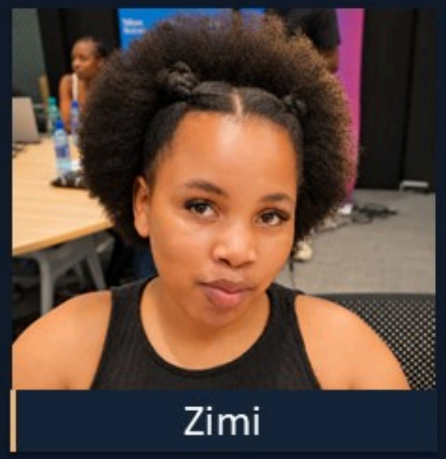

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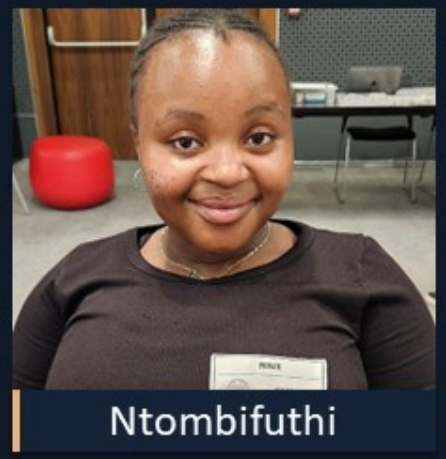

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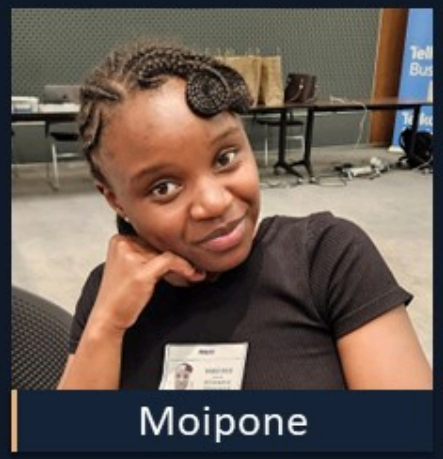

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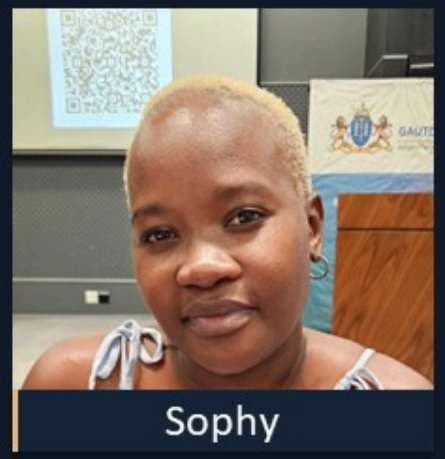

Cebolihle


Zimi


Ntombifuthi


Sinelizwe


Moipone


Sophy

PROBLEM · SOLUTION · PILOT PLAN

SA Educate

Breaking communication barriers. Building equal educational opportunities.

A digital learning concept for Deaf and hard-of-hearing learners in South Africa

THE PROBLEM

Lessons that aren’t accessible leave Deaf learners behind

Many educators can’t sign, and qualified interpreters and accessible resources are limited. Learners struggle to follow lessons, ask questions and show what they know, which leads to unequal learning experiences.

2,136
learners in Gauteng’s special schools for the Deaf (2017), second only to KwaZulu-Natal

7
Gauteng schools cater specifically for hearing-impaired learners

2023
the year SASL became an official language, but rollout is under-resourced

THE SOLUTION

A platform that bridges learners and educators

Speech-to-sign-language support	Turns spoken lessons into sign-language interpretation.
Sign-language-to-text support	Deaf learners ask and answer questions in text.
Interactive learning support	Fuller participation in classroom activities.
AI-powered educational assistance	Helps educators with lesson prep and curriculum information.
Accessible learning resources	Content designed for Deaf and hard-of-hearing learners.

HOW IT WORKS

Three steps from speech to sign

1 Teacher speaks

The teacher asks a question or explains a lesson.

→

2 Speech is processed

Speech recognition captures the message and prepares it for translation.

→

3 Interpretation shown

A qualified human interpreter or a validated signing solution presents the message.

Accuracy comes first. SASL has its own grammar. Subtitles alone are not sign-language interpretation, and automated signing is not presented as reliable until validated with Deaf users and qualified SASL interpreters.

IMPACT

Impact on Gauteng communities

- Gives learners with disabilities equal opportunities to access education.
- Reduces dependence on teachers, parents and assistants for basic digital interactions.
- Improves access to learning materials and assessments.
- Helps learners feel capable and included in the learning process.
- Makes digital education available through different communication methods.
- Helps bridge the digital divide for learners with disabilities.

Funding

- Department of Education
- Private stakeholders

NEXT STEPS

Start with a focused classroom pilot

1. Engage Deaf learners, teachers and qualified SASL interpreters in the design.
2. Identify the most common classroom situations and phrases for the first pilot.
3. Build a prototype of the speech-to-interpretation workflow.
4. Test accuracy, clarity, usability and accessibility with Deaf users and interpreters.
5. Collect feedback, measure participation and refine before expanding.

Every learner deserves to understand, participate and succeed, regardless of their hearing ability.

Thank you.